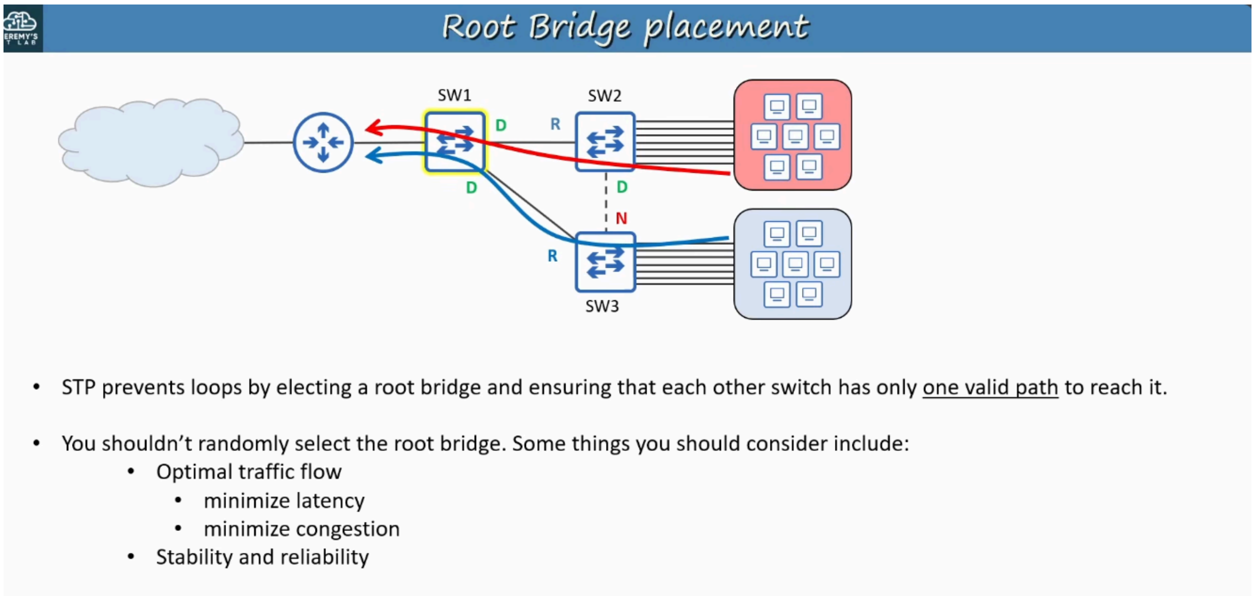
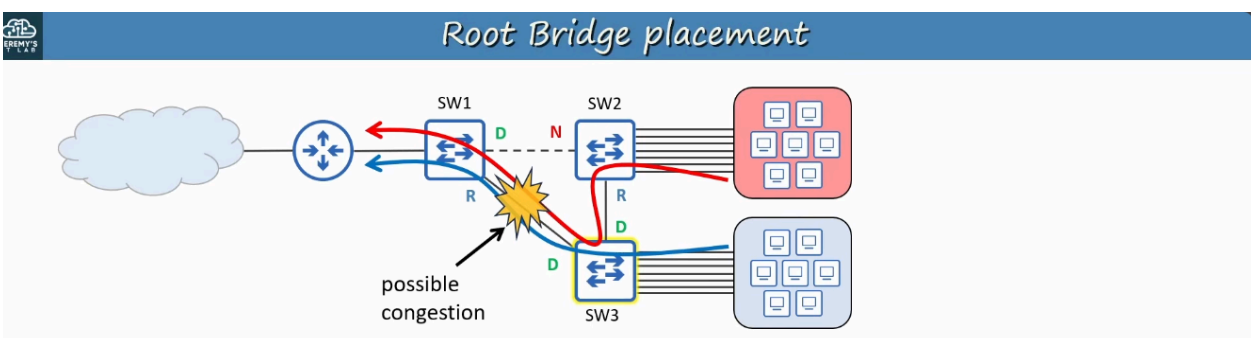
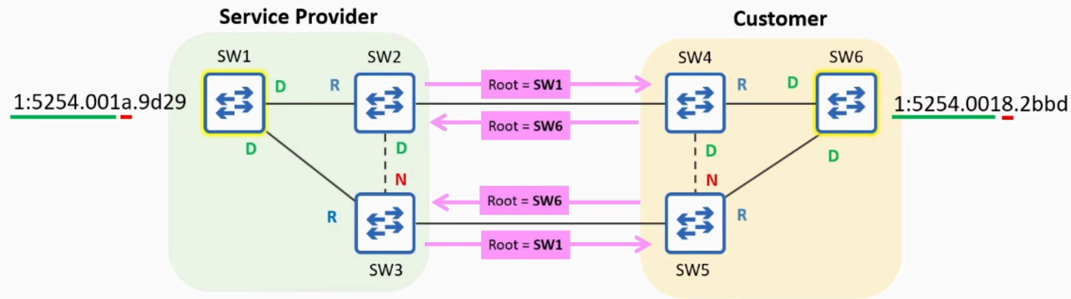
Day 21 (Part 5): Root Guard (STP Toolkit)



If SW3 takes the Root Bridge



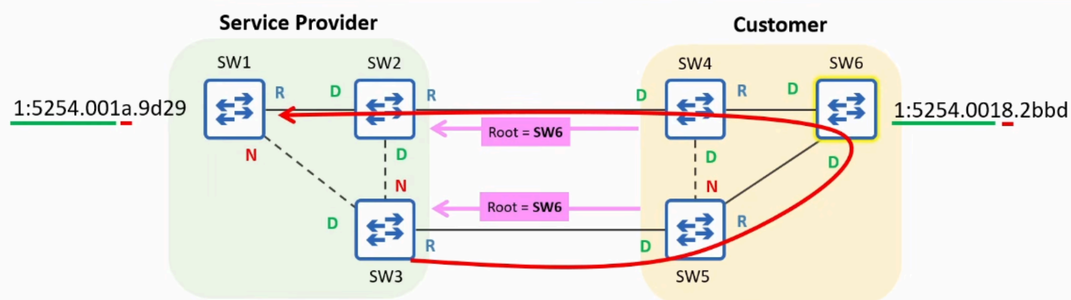
Root Guard: the problem



- Within your own LAN, you can easily control the root bridge by setting its priority to **0**.
  - But there are cases where you might connect your LAN to other switches outside of your direct control:
    - A service provider offering **Metro Ethernet** service to customers
      - often used to connect sites within a MAN (Metropolitan Area Network)

- Even if you set your root bridge's priority to 0, its role can be taken by another switch with a lower MAC address.

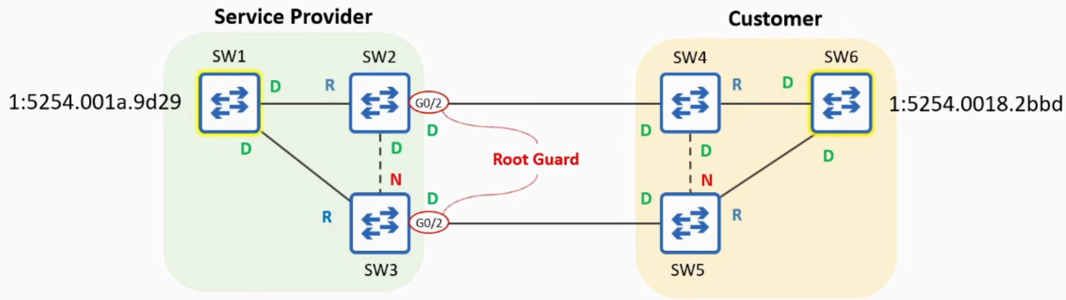
Root Guard: the problem



- With no safeguard in place, SW1, SW2, and SW3 accept SW6 as the root bridge, affecting the service provider's STP topology.
  - Frames from SW3 to SW1 must take a detour through the customer's LAN.

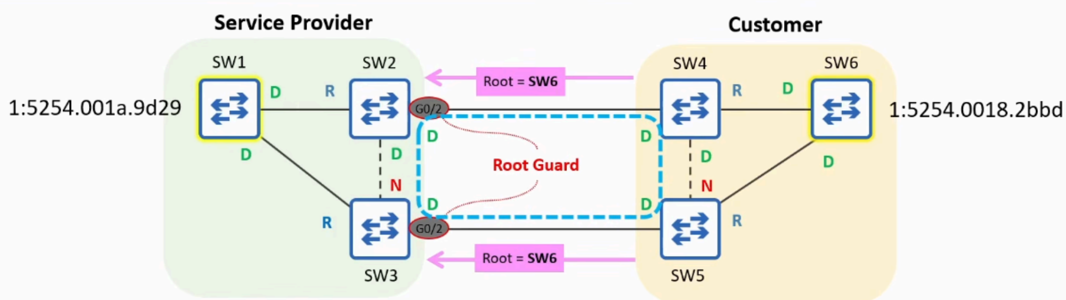
- **Root Guard** can be configured to protect your STP topology by preventing your switches from accepting **superior BPDUs** from switches outside of your control.
  - **Superior BPDU** = a BPDU that is superior in the STP algorithm (e.g. claiming a better root bridge ID).

Root Guard: the solution



- If you want to ensure that the root bridge remains in your LAN, you can configure **Root Guard** on the ports connected to switches outside of your control.
  - SW2 G0/2, SW3 G0/2
- Use `SW2(config-if)#` **`spanning-tree guard root`** to enable Root Guard on a port.
  - There is no command to enable it by default from global config mode.

Root Guard: the solution



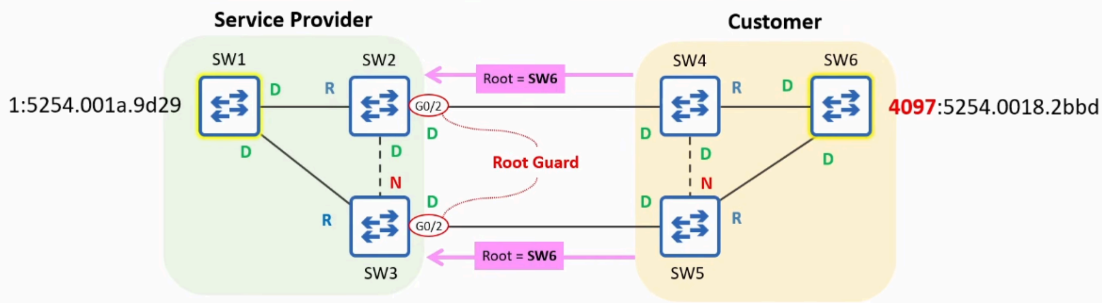
- If you want to ensure that the root bridge remains in your LAN, you can configure **Root Guard** on the ports connected to switches outside of your control.
  - SW2 G0/2, SW3 G0/2
- Use `SW2(config-if)#` **`spanning-tree guard root`** to enable Root Guard on a port.
  - There is no command to enable it by default from global config mode.
- If a Root Guard-enabled port receives a BPDU, it will enter the **Broken** (*Root Inconsistent)* state, effectively disabling it.
  - The port will not be able to forward data frames and will discard any frames it receives.
  - SW1, SW2, and SW3 won't accept SW6 as the root bridge.

In this special case, SW2 - SW3 - SW4 - SW5 will have 2 DESIGNATED PORT

REMEMBER:

- There should be 1 DESIGNATED PORT per link in STP

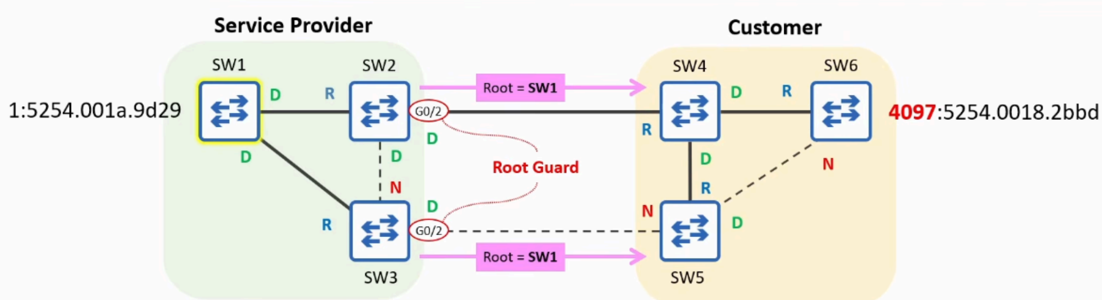
Root Guard: the solution

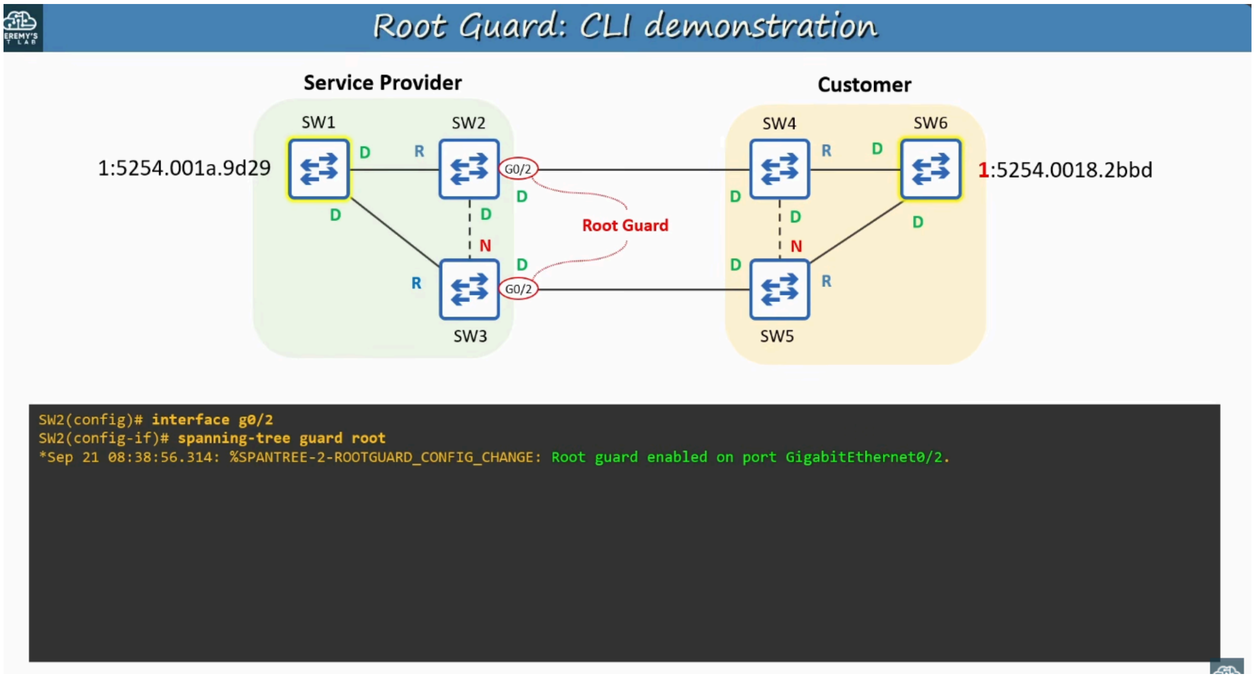


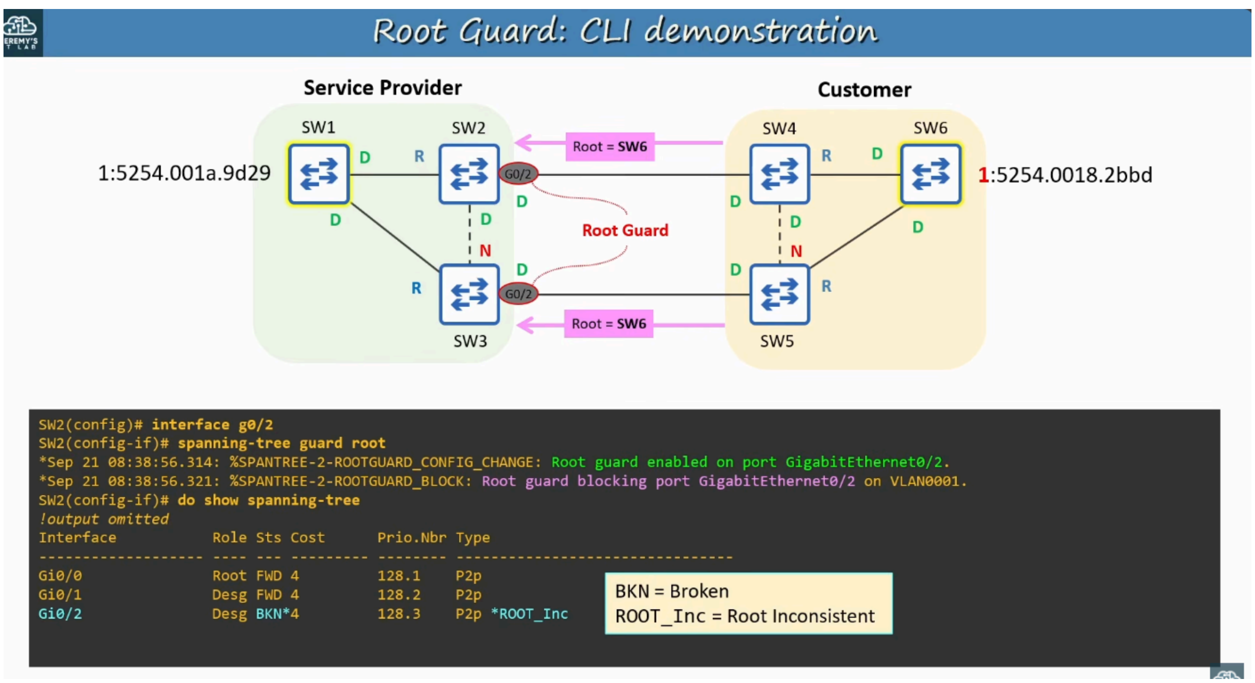
- To re-enable a port disabled by Root Guard, you must solve the issue that disabled the port.
  - The disabled port must stop receiving superior BPDUs.
  - Tell the customer to increase the priority value of their switch.
- Once the superior BPDUs received by SW2 G0/2 and SW3 G0/3 age out, the ports will automatically be re-enabled.
  - A BPDU's Max Age is 20 seconds by default.

Then CUSTOMER's LAN will take SW1 as a Root Bridge

Root Guard: the solution

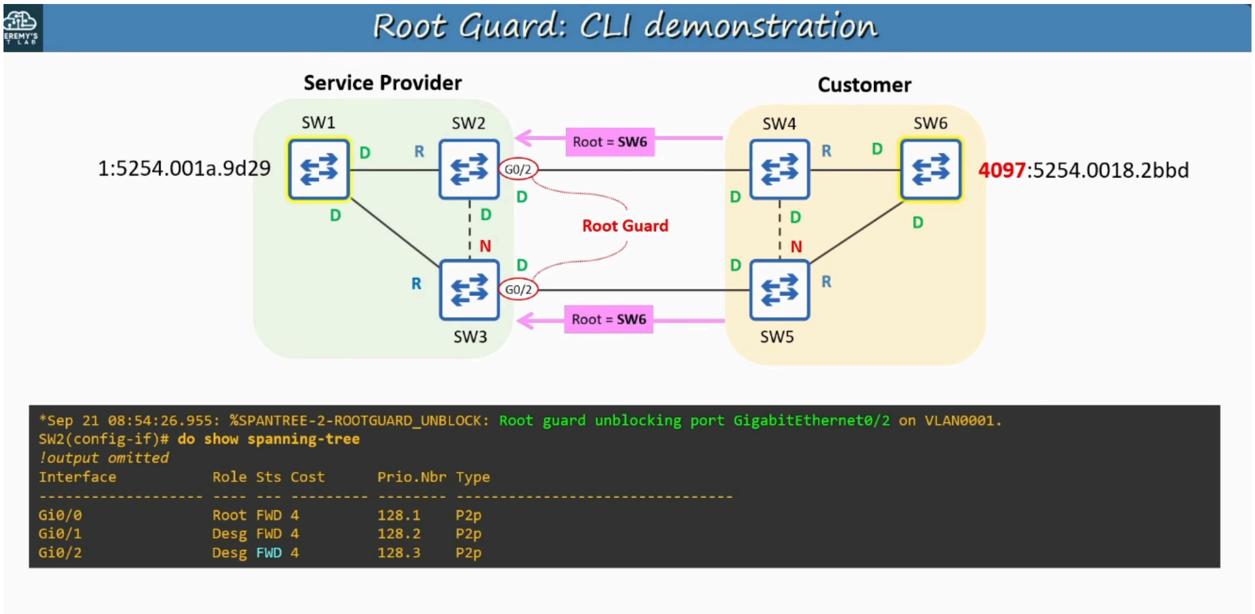




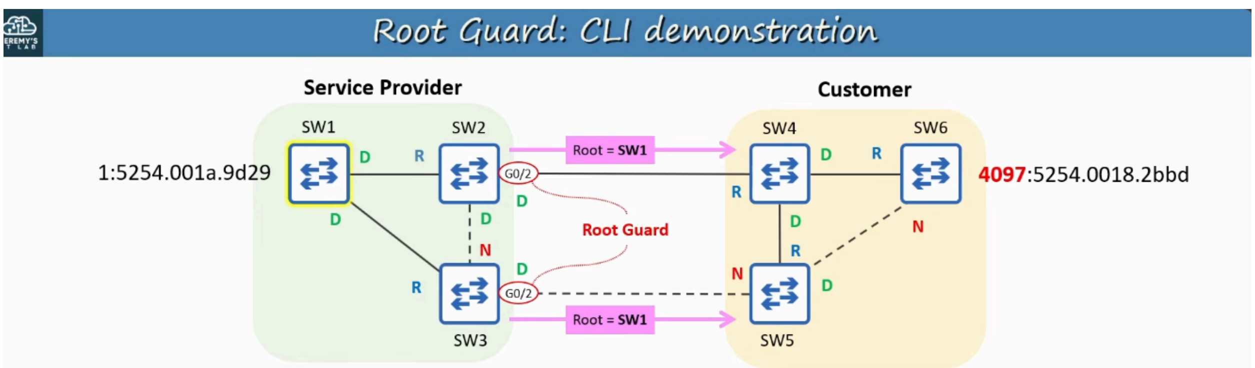


- Broken: can not forward or receive data frame
- Root inconsistent: tell us why port is broken

Change SW6 priority to 4097 and after 20 SECONDS, the interface is unblocked



Then Customer's LAN will take SW1 as Root Bridge



Summary

- When selecting a LAN's root bridge, you should consider the following:
  - Optimal traffic flow
    - minimize latency
    - minimize congestion
  - Stability and reliability
- Within your own LAN, you can easily control the root bridge by setting its priority to 0.
  - There are cases where you might connect your switches to other switches outside of your control (e.g. service provider + client).
- **Root Guard** can be configured on specific ports to prevent them from accepting **superior BPDUs** from switches outside of your control.
- Use `SW(config-if)#` **`spanning-tree guard root`** to enable Root Guard on a port.
  - There is no command to enable it by default from global config mode.
- Root Guard prevents a port from becoming a root port if it receives a superior BPDU.
  - If the port receives a superior BPDU, it becomes Broken (BKN) / Root Inconsistent (`ROOT_Inc`).
- If the port stops receiving superior BPDUs, it will automatically recover.